ECS7032P Assignment 1 – Individual Reflection

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Group report: Decoupled Information-Set MCTS with a Card-Value Model for Sushi Go!

We split the work equally three ways, and my part was the code. I set up TAG and our package and built the agent as a modified copy of BasicMCTS within the rules: nothing from the advanced MCTS agent, no change to the framework, everything inside the groupW package. As our report states, we wrote the code with Claude; my part was to direct that work, turn the team's design into concrete modifications (card-value rollouts, round-end rollouts, max-n backups, decoupled selection over information sets and separate four-player settings), keep each one as a switch and test it. I oversaw the experiments behind the report, 167 runs, and checked their results. I also prepared the competition file for the 1-second limit, with a wall-clock guard, and checked that the agent compiles and plays in a fresh copy of TAG.

For what to build I relied on Weronika's reading of the literature, and for what to test and when, on Hakan's plan and his decisions as group leader.

Building the agent taught me that the evaluation matters more than the details of the search: without our heuristic the agent lost 28.3 percentage points, more than any change to the search gained. I also learned that more thinking time does not automatically help: our lead shrank at 500 ms while TAG's ISMCTS gained. Next time I would test at the competition's time limit earlier.

Use of generative AI: as our report states, we used Claude (Anthropic) for the code and the drafting; this reflection was also drafted with Claude from our own account of the work.

Word count: 263 (without the heading)